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Mixtures of hard spheres in the Percus-Yevick approximation. Light scattering at finite angles

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In a previous paper (I), a new equation for the light scattering (or small angle x-ray or neutron scattering) of a concentrated p -component mixture of spherical (colloidal) particles in a low-molecular weight solvent was derived. Use was made of Baxter's factorization of the direct correlation matrix. It was found that the light scattering intensity can be formulated in factorized form as well. The formalism was applied to a multicomponent system of hard spheres treated in the Percus-Yevick approximation. For zero scattering angle, a rather simple, exact expression was obtained. In this paper it is proved that a closed expression can also be obtained for finite scattering angles. It contains at most 18 (averaged) functions of the scattering angle for any number of hard sphere components. This makes it possible to apply the equation to a continuous distribution of hard sphere diameters. A series expansion is given for small scattering wave numbers.

I. INTRODUCTION

In a previous paper,¹ further denoted by I, we derived a new equation for the intensity of waves scattered by a multicomponent (fluid) mixture of particles with spherically symmetric interparticle interactions in terms of Q functions introduced by Baxter.² Often these Q functions have a simpler mathematical structure than the well known total correlation functions $h(r)$ and the direct correlation functions $c(r)$, although their physical meaning is less straightforward. They are of particular importance in the analysis of *multicomponent* systems where the other functions are unwieldy or unknown. Examples are the Percus-Yevick approximation for hard spheres,^{2(b)} for hard spheres with surface adhesion,^{3(a),(b)} charged hard spheres,^{3(c)} and other hard-sphere models.^{3(d)} The application of the hard-sphere (HS) model seems of particular importance for the broad field of spherical macromolecules and colloidal particles.⁴

In paper I, we found that the intensity of scattered light or other types of radiation as a function of the scattering vector K can be directly expressed in determinants containing as elements certain Fourier transforms $Q_{ik}(K)$ belonging to the pair of particle species i, k . Further, it was found that, for hard spheres treated in the Percus-Yevick approximation, these determinants can be evaluated exactly at $K=0$ for any number of HS components, and that the scattering intensity can be simply expressed in terms of mean products of HS diameters and scattering amplitudes.

In this paper, we will show that this can also be proved for $K \neq 0$. The resulting equations can be used, for example, to study the distortion of the structure factor of a concentrated dispersion of colloidal spheres caused by polydispersity in the particle size. A series expansion will be derived up to and including terms in K^2 .

II. SCATTERING EQUATIONS

Let us recapitulate some relevant equations from paper I. For a p -component mixture of particles with spherically symmetric interparticle interactions, Fourier⁵ derived

$$R(K) = \sum_{i=1}^p f_i^2 \rho_i \langle B_i^2 \rangle_a - \langle B_i \rangle_a^2 + \sum_{i,k=1}^p f_i f_k \langle B_i \rangle_a \langle B_k \rangle_a (\rho_i \rho_k)^{1/2} [\delta_{ik} + \tilde{H}_{ik}(K)] . \quad (1)$$

Here, $R(K)$ is the normalized scattering intensity (Rayleigh ratio), ρ_i is the number density, f_i is the scattering amplitude at zero angle, and B_i is the intraparticle interference factor of particles i ; δ_{ik} is the Kronecker delta ($\delta_{ik}=0$ for $i \neq k$ and $\delta_{ik}=1$ for $i=k$). Further,

$$\tilde{H}_{ik}(K) = (\rho_i \rho_k)^{1/2} \int_0^\infty 4\pi r^2 h_{ik}(r) [(Kr)^{-1} \sin(Kr)] dr , \quad (2)$$

where $h_{ik}(r) = g_{ik}(r) - 1$ is the total correlation function and $g_{ik}(r)$ the radial distribution function as a function of r , the distance between the centers of the interparticle interaction of species i and k . Further, $K = (4\pi/\lambda) \times \sin(\theta/2)$, with θ the scattering angle and λ the wavelength of the radiation. The brackets $\langle \rangle_a$ indicate that angular averages are taken. The first sum in Eq. (1) is often called the "incoherent part" and the second sum the "coherent part." In our further discussion, we will only consider the coherent part and drop the angular brackets for convenience.

The expression $\delta_{ik} + \tilde{H}_{ik}(K)$ is sometimes called⁶ the partial structure factor with the symbol $S_{ik}(K)$. In the case of light scattering of colloidal particles dispersed in a low-molecular liquid (in the so-called Rayleigh-Gans-Debye approximation⁷), $f_i = 2^{1/2} \pi n \lambda_0^{-2} \gamma_i$, where n is the refractive index of the colloidal dispersion, λ_0 is the wavelength of the incident light *in vacuo* ($\lambda = \lambda_0/n$), and γ_i is the refractive index increment $\partial n / \partial \rho_i$ of species i .

Another representation of the scattering is possible with the use of $\tilde{C}_{ik}(K)$, the Fourier transform of the direct correlation function $c_{ik}(r)$, first introduced by Ornstein and Zernike⁸ for a one-component system and extended to multicomponent systems by Pearson and Rushbrooke⁹ and Lebowitz.¹⁰ The functions $\tilde{C}_{ik}(K)$ are defined by the following matrix equation:

$$[\mathbf{E} + \tilde{\mathbf{H}}(K)][\mathbf{E} - \tilde{\mathbf{C}}(K)] = \mathbf{E} , \quad (3)$$

where $\tilde{\mathbf{H}}(K)$ and $\tilde{\mathbf{C}}(K)$ are $p \times p$ matrices with elements $\tilde{H}_{ik}(K)$ defined by Eq. (2) and $\tilde{C}_{ik}(K)$ defined by a similar equation with $h_{ik}(r)$ replaced by $c_{ik}(r)$. Further, $\mathbf{E} = [\delta_{ik}]$ is the unit matrix. This gives

$$\delta_{ik} + \tilde{H}_{ik}(K) = S_{ik}(K) = [\Delta(K)]^{-1} |\mathbf{E} - \tilde{\mathbf{C}}(K)|^{ki}, \quad (4)$$

where

$$\Delta(K) = |\mathbf{E} - \tilde{\mathbf{C}}(K)| = |\delta_{ik} - \tilde{C}_{ik}(K)| \quad (5)$$

is the $p \times p$ determinant of the matrix $\mathbf{E} - \tilde{\mathbf{C}}(K)$ and $|\mathbf{E} - \tilde{\mathbf{C}}(K)|^{ki}$ is the cofactor of the element $\delta_{ki} - \tilde{C}_{ki}(K)$ of this determinant. The scattering then becomes

$$R(K) = [-D_f(K)] [\Delta(K)]^{-1}, \quad (6)$$

where the determinant $D_f(K)$ is given by the $(p+1) \times (p+1)$ determinant

$$D_f(K) = \begin{vmatrix} 0 & f_k B_k \rho_k^{1/2} \\ f_i B_i \rho_i^{1/2} & \delta_{ik} - \tilde{C}_{ik}(K) \end{vmatrix}. \quad (7)$$

Note that D_f and D as defined in paper I differ by a constant factor.

Because the $c_{ik}(r)$ are often theoretically more accessible than the $h_{ik}(r)$, Eq. (7) may have a greater utility than Eq. (1). For instance, Lebowitz¹⁰ was able to generalize the Wertheim^{11(a)} and Thiele^{11(b)} solution of the PY equation for hard spheres to a multicomponent system and to express the $c_{ik}(r)$ for a two-component mixture in terms of polynomials in r . This result was subsequently applied by Ashcroft and Langreth⁵ to calculate the scattering of a two-component mixture of hard spheres.

It was Baxter^{2(b)} who showed that, when the direct correlation functions $c_{ik}(r)$ have a finite range so that for a chosen set of range parameters $d_{11}, \dots, d_{pp}$ and associated quantities $d_{ik} = \frac{1}{2}(d_{ii} + d_{kk})$ and $s_{ik} = \frac{1}{2}(d_{ii} - d_{kk})$ the $c_{ik}(r)$ are zero for $r > d_{ik}$, the direct correlation matrix $\mathbf{E} - \tilde{\mathbf{C}}(K)$ may be factored as follows:

$$\mathbf{E} - \tilde{\mathbf{C}}(K) = \tilde{\mathbf{Q}}^T(-K) \tilde{\mathbf{Q}}(K), \quad (8)$$

where the elements of $\tilde{\mathbf{Q}}(K)$ are defined in terms of the auxiliary function $q_{ik}(r)$:

$$\tilde{Q}_{ik}(K) = \delta_{ik} - 2\pi(\rho_i \rho_k)^{1/2} \int_{s_{ik}}^{d_{ik}} dr q_{ik}(r) e^{iKr}. \quad (9)$$

The real advantage is that in several cases,³ e.g., the PY approximation for hard spheres [see Eq. (14) below], the $q_{ik}(r)$ and $\tilde{Q}_{ik}(K)$ have a much simpler mathematical representation than the $c_{ik}(r)$ and $\tilde{C}_{ik}(K)$.

In paper I, we showed that Eqs. (5)–(7) for the (light) scattering intensity can also be factored, as

$$-D_f(K) = \sum_{j=1}^p L_j(K) L_j(-K), \quad (10)$$

$$\Delta(K) = |\tilde{\mathbf{Q}}^T(-K)| |\tilde{\mathbf{Q}}(K)|, \quad (11)$$

with

$$L_j(K) = \sum_{i=1}^p (f_i B_i \rho_i^{1/2}) |\tilde{\mathbf{Q}}(K)|^{ji}. \quad (12)$$

The $L_j(K)$ can also be written as a single determinant [see Eq. (25) of paper I]. Note that also the L_j used

here differ by a constant factor from those used in paper I. The partial structure factors are given by

$$S_{ik}(K) = [\Delta(K)]^{-1} \sum_{j=1}^p |\tilde{\mathbf{Q}}(-K)|^{jk} |\tilde{\mathbf{Q}}(K)|^{ji}. \quad (13)$$

Light scattering of hard spheres: For a p -component mixture of hard spheres with diameters d_{ii} , $q_{ik}(r)$ in the PY approximation is given by^{2(b)}

$$q_{ik}(r) = \frac{1}{2} a_i (r^2 - d_{ik}^2) + b_i (r - d_{ik}) \quad (14)$$

for $s_{ik} < r < d_{ik}$, where

$$a_i = (1 - \xi_3)^{-2} (1 - \xi_3 + 3\xi_2 d_{ii}), \quad (15)$$

$$b_i = -\frac{3}{2} d_{ii}^2 (1 - \xi_3)^{-2} \xi_2, \quad (16)$$

and

$$\xi_\nu = \frac{1}{6} \pi \sum_{j=1}^p \rho_j d_{jj}^\nu, \quad (17)$$

$$d_{ik} = \frac{1}{2} (d_{ii} + d_{kk}), \quad s_{ik} = \frac{1}{2} (d_{ii} - d_{kk}). \quad (18)$$

For $K=0$,

$$\tilde{Q}_{ik}(0) = \delta_{ik} + (\rho_i \rho_k)^{1/2} Z_{ik}(0), \quad (19)$$

$$Z_{ik}(0) = \pi d_{kk}^2 (1 - \xi_3)^{-2} \left[\frac{1}{3} (1 - \xi_3) (2d_{ii} - s_{ik}) + \frac{1}{2} \xi_2 d_{ii} d_{kk} \right]. \quad (20)$$

When Eqs. (19) and (20) are substituted into $|\tilde{\mathbf{Q}}(0)|$, we find that this determinant has rank 2. Then it can be shown that

$$\Delta(0) = [(1 + 2\xi_3)(1 - \xi_3)^{-2}]^p, \quad (21)$$

and that, for example, for $f_i = \alpha_3 d_{ii}^3$,

$$-D_f(0) = (6/\pi) \alpha_3^2 (1 - \xi_3)^{-4} \left[(1 + 2\xi_3)^2 (\xi_6 + \xi_3^2 \xi_6 - 2\xi_3 \xi_6) + 9\xi_4^3 (1 - \xi_3)^2 - 6(1 + 2\xi_3) \xi_4 \xi_5 (1 - \xi_3)^2 \right], \quad (22)$$

where $\alpha_3 = 2^{1/2} \pi n \lambda_0^{-2} (\pi/6) dn/d\xi_3$. One observes that the scattering for $f_i \propto d_{ii}^3$ is characterized by only four averages ξ_3, ξ_4, ξ_5 , and ξ_6 . For the more general case, see paper I.

III. SCATTERING OF HARD SPHERES AT FINITE ANGLES

To obtain $\tilde{Q}_{ik}(K)$ for $K \neq 0$, Eq. (14) must be substituted into Eq. (9). With straightforward algebra, one finds that Eq. (9) can be cast into the following form:

$$\tilde{Q}_{ik}(K) = \delta_{ik} + (\rho_i \rho_k)^{1/2} Z_{ik}(K), \quad (23)$$

$$Z_{ik}(K) = e^{iX_k} (M_k + d_{ii} N_k), \quad (24)$$

where M_k and N_k are functions of d_{kk} :

$$M_k = (\pi/6) (1 - \xi_3)^{-1} d_{kk}^3 \Phi(X_k), \quad (25)$$

$$N_k = (\pi/6) (1 - \xi_3)^{-1} d_{kk}^2 \times [3\Psi(X_k) - iX_k \Phi(X_k) + 3(1 - \xi_3)^{-1} \xi_2 d_{kk} \Phi(X_k)], \quad (26)$$

where $X_k = \frac{1}{2} K d_{kk}$, and

$$\Psi(X_k) = X_k^{-1} \sin X_k; \quad \Phi(X_k) = 3X_k^{-3} (\sin X_k - X_k \cos X_k). \quad (27)$$

Here, i —when it is not a subscript—is the imaginary unit. Note that $Z_{ik}(K) \neq Z_{ki}(K)$ and that $Z_{ik}(-K)$ is the complex conjugate of $Z_{ik}(K)$. If desired, the functions Ψ and Φ can also be expressed as spherical Bessel func-

tions: $\Psi(X) = j_0(X)$; $\Phi(X) = 3X^{-1}j_2(X)$. For $K=0$, Eqs. (23) and (24) reduce to Eqs. (19) and (20).

Making up determinants with elements $Z_{ik}(K)$ as given by Eq. (24), one observes that the exponentials can be factored out and that further reduction is achieved by subtractions¹² of rows and columns. Then it is easily verified that the 2×2 determinant

$$\begin{vmatrix} Z_{ap} & Z_{aq} \\ Z_{bp} & Z_{bq} \end{vmatrix} = e^{iX_a} e^{iX_b} (d_{aa} - d_{bb}) (N_p M_q - M_p N_q), \quad (28)$$

and that all determinants of this type which are of higher order are zero. This implies that the $p \times p$ matrices $\mathbf{Z}(K)$ have rank 2. Then it follows (see paper I) that

$$|\tilde{\mathbf{Q}}(K)| = 1 + \sum_{i=1}^p \rho_i Z_{ii} + \frac{1}{2!} \sum_{i,j=1}^p \rho_i \rho_j \begin{vmatrix} Z_{ii} & Z_{ij} \\ Z_{ji} & Z_{jj} \end{vmatrix}, \quad (29)$$

which can be cast in the following form

$$|\tilde{\mathbf{Q}}(K)| = 1 + (6/\pi) \langle M e^{iX} \rangle + (6/\pi) \langle d N e^{iX} \rangle + (6/\pi)^2 (\langle M e^{iX} \rangle \langle d N e^{iX} \rangle - \langle d M e^{iX} \rangle \langle N e^{iX} \rangle), \quad (30)$$

where, as a shorthand notations, the following symbolic expressions are introduced for the *unnormalized* averages ($\nu = 0, 1$):

$$\begin{aligned} \langle d^\nu M e^{iX} \rangle &\equiv (\pi/6) \sum_{k=1}^p \rho_k d_{kk}^\nu M_k e^{iX_k}; \\ \langle d^\nu N e^{iX} \rangle &\equiv (\pi/6) \sum_{k=1}^p \rho_k d_{kk}^\nu N_k e^{iX_k}. \end{aligned} \quad (31)$$

Substituting the expressions (25) and (26) for M_k and N_k , respectively, one finds that

$$\begin{aligned} (1 - \xi_3)^2 [1 + (6/\pi) \langle M e^{iX} \rangle + (6/\pi) \langle d N e^{iX} \rangle] \\ = (1 - \xi_3)^2 + (1 - \xi_3) \langle d^3 e^{iX} \Phi \rangle + 3(1 - \xi_3) \langle d^3 e^{iX} \Psi \rangle \end{aligned}$$

$$|\mathbf{Q}(K)|^{jj} = |\mathbf{Q}(K)| \delta_{ij} - (\rho_i \rho_j)^{1/2} Z_{ij} - (\rho_i \rho_j)^{1/2} \sum_{k=1}^p \rho_k \begin{vmatrix} Z_{ij} & Z_{ik} \\ Z_{kj} & Z_{kk} \end{vmatrix}, \quad (40)$$

one finds that $L_j(K)$ can be written as follows:

$$\begin{aligned} L_j(K) &= \rho_j^{1/2} f_j B_j(K) |\mathbf{Q}(K)| + (6/\pi) \rho_j^{1/2} M_j G_I(K) \\ &\quad + (6/\pi) \rho_j^{1/2} N_j G_{II}(K), \end{aligned} \quad (41)$$

where

$$\begin{aligned} G_I(K) &= -\langle f B e^{iX} \rangle + 3(1 - \xi_3)^{-1} U_I(K) \\ &\quad + (1 - \xi_3)^{-2} [3\xi_2 - \frac{1}{2} iK(1 - \xi_3)] U_{II}(K), \end{aligned} \quad (42)$$

$$G_{II}(K) = -\langle d f B e^{iX} \rangle - (1 - \xi_3)^{-1} U_{II}(K), \quad (43)$$

with

$$U_I(K) = \langle d f B e^{iX} \rangle \langle d^2 e^{iX} \Psi \rangle - \langle f B e^{iX} \rangle \langle d^3 e^{iX} \Psi \rangle, \quad (44)$$

$$U_{II}(K) = \langle d f B e^{iX} \rangle \langle d^3 e^{iX} \Phi \rangle - \langle f B e^{iX} \rangle \langle d^4 e^{iX} \Phi \rangle, \quad (45)$$

Substituting Eqs. (25) and (26) for M_k and N_k , respectively, and rearranging terms, we obtain finally the following rather simple equation for $L_j(K)$:

$$-\frac{1}{2} (1 - \xi_3) iK \langle d^4 e^{iX} \Phi \rangle + 3\xi_2 \langle d^4 e^{iX} \Phi \rangle, \quad (33)$$

and that

$$\begin{aligned} (1 - \xi_3)^2 (6/\pi)^2 (\langle M e^{iX} \rangle \langle d N e^{iX} \rangle - \langle d M e^{iX} \rangle \langle N e^{iX} \rangle) \\ = 3 \langle d^3 e^{iX} \Psi \rangle \langle d^3 e^{iX} \Phi \rangle - 3 \langle d^2 e^{iX} \Psi \rangle \langle d^4 e^{iX} \Phi \rangle, \end{aligned} \quad (34)$$

where, analogously as before, we define

$$\langle d^\nu e^{iX} \Psi \rangle \equiv (\pi/6) \sum_{k=1}^p \rho_k d_{kk}^\nu e^{iX_k} \Psi(X_k), \quad (35)$$

$$\langle d^\nu e^{iX} \Phi \rangle \equiv (\pi/6) \sum_{k=1}^p \rho_k d_{kk}^\nu e^{iX_k} \Phi(X_k). \quad (36)$$

With these expressions we obtain

$$(1 - \xi_3)^2 |\tilde{\mathbf{Q}}(K)| = F_{11} F_{22} - F_{12} F_{21}, \quad (37)$$

with

$$\begin{aligned} F_{11}(K) &= 1 - \xi_3 + \langle d^3 e^{iX} \Phi \rangle, \\ F_{12}(K) &= \langle d^4 e^{iX} \Phi \rangle, \\ F_{22}(K) &= 1 - \xi_3 + 3 \langle d^3 e^{iX} \Psi \rangle, \\ F_{21}(K) &= \frac{1}{2} (1 - \xi_3) iK - 3\xi_2 + 3 \langle d^2 e^{iX} \Psi \rangle. \end{aligned} \quad (38)$$

Further, $|\tilde{\mathbf{Q}}^T(-K)| = |\tilde{\mathbf{Q}}(-K)|$ is obtained simply by replacing K in $|\tilde{\mathbf{Q}}(K)|$ by $-K$; it is also the complex conjugate of $|\tilde{\mathbf{Q}}(K)|$. Then, $\Delta(K)$ follows from Eq. (11):

$$\Delta(K) = (1 - \xi_3)^{-4} [(F_{11} F_{22} - F_{12} F_{21})(F_{11}^* F_{22}^* - F_{12}^* F_{21}^*)], \quad (39)$$

where $F_{\alpha\beta}^*$ is the complex conjugate of $F_{\alpha\beta}$ [or $F_{\alpha\beta}^*(K) = F_{\alpha\beta}(-K)$].

Determinant $D_f(K)$: The determinant $D_f(K)$ can be obtained in a similar way from the $L_j(K)$ [see Eq. (12)]. This can be done as follows: By making use of the equations (see paper I)

$$\rho_j^{-1/2} (1 - \xi_3)^2 L_j(K) = \begin{vmatrix} f_j B_j & \langle f B e^{iX} \rangle & \langle d f B e^{iX} \rangle \\ d_{jj}^3 \Phi_j & F_{11} & F_{12} \\ 3d_{jj}^2 \Psi_j & F_{21} & F_{22} \end{vmatrix}. \quad (46)$$

To perform the summation over j , we write

$$(1 - \xi_3)^2 L_j(K) = \rho_j^{1/2} f_j B_j T_1 + \rho_j^{1/2} d_{jj}^3 \Phi_j T_2 + 3\rho_j^{1/2} d_{jj}^2 \Psi_j T_3, \quad (47)$$

with

$$\begin{aligned} T_1 &= F_{11} F_{22} - F_{12} F_{21}, \\ T_2 &= F_{21} \langle d f B e^{iX} \rangle - F_{22} \langle f B e^{iX} \rangle, \\ T_3 &= F_{12} \langle f B e^{iX} \rangle - F_{11} \langle d f B e^{iX} \rangle. \end{aligned} \quad (48)$$

Then one finds, with Eq. (10),

$$\begin{aligned} -(\pi/6) (1 - \xi_3)^4 D_f(K) &= \langle f^2 B^2 \rangle T_1 T_1^* + \langle d^6 \Phi^2 \rangle T_2 T_2^* \\ &\quad + 9 \langle d^4 \Psi^2 \rangle T_3 T_3^* + \langle f B d^3 \Phi \rangle (T_1 T_2^* + T_1^* T_2) + 3 \langle f B d^2 \Psi \rangle \\ &\quad \times (T_1 T_3^* + T_1^* T_3) + 3 \langle d^5 \Phi \Psi \rangle (T_2 T_3^* + T_2^* T_3), \end{aligned} \quad (49)$$

which together with Eqs. (39) and (6) lead to the scattering intensity $R(K)$. In the next section, a series expansion of $R(K)$ up to and including terms in K^2 will be given.

IV. SERIES EXPANSION IN K^2

We consider the special case of a centrosymmetric distribution of scattering material in the particles with center coinciding with HS center, and a scattering amplitude proportional to the HS volume; thus,

$$f_j/\alpha_j = d_{jj}^3, \quad (50)$$

$$B_j = 1 - K^2 \Gamma d_{jj}^2, \quad (51)$$

For homogeneous spheres, for example, $\Gamma = 1/40$. The relevant quantities in Eqs. (48) and (49) then become, up to and including terms in K^2 ,

$$\langle f^2 B^2 \rangle \alpha_3^{-2} = \xi_6 - 2K^2 \Gamma \xi_8, \quad (52)$$

$$\langle d^6 \Phi^2 \rangle = \xi_6 - \frac{1}{20} K^2 \xi_8, \quad (53)$$

$$\langle d^4 \Psi^2 \rangle = \xi_4 - \frac{1}{12} K^2 \xi_6, \quad (54)$$

$$\langle f B d^3 \Phi \rangle \alpha_3^{-1} = \xi_6 - K^2 \xi_8 (\Gamma + \frac{1}{40}), \quad (55)$$

$$\langle f B d^2 \Phi \rangle \alpha_3^{-1} = \xi_5 - K^2 \xi_7 (\Gamma + \frac{1}{24}), \quad (56)$$

$$\langle d^5 \Phi \Psi \rangle = \xi_5 - \frac{1}{15} K^2 \xi_7, \quad (57)$$

$$\langle f B e^{iX} \rangle \alpha_3^{-1} = \xi_3 + \frac{1}{2} i K \xi_4 - K^2 \xi_5 (\Gamma + \frac{1}{8}), \quad (58)$$

$$\langle d f B e^{iX} \rangle \alpha_3^{-1} = \xi_4 + \frac{1}{2} i K \xi_5 - K^2 \xi_6 (\Gamma + \frac{1}{8}). \quad (59)$$

After performing tedious algebra, we find

$$R(K) = \alpha_3^2 (6/\pi) \frac{[Y_3 - K^2 (\Gamma Y_4 + \frac{1}{20} Y_5)]}{(Y_1 - K^2 Y_2)}, \quad (60)$$

where

$$Y_1 = (1 + 2\xi_3)^2, \quad (61)$$

$$Y_2 = -\frac{1}{5} \xi_5 (1 + 2\xi_3)^2 + (1 + 2\xi_3) (\frac{1}{2} \xi_4^2 + \xi_5) - \frac{9}{4} \xi_4^2, \quad (62)$$

$$Y_3 = (1 - \xi_3)^2 [\xi_6 (1 + 2\xi_3)^2 + 9\xi_4^3 - 6\xi_4 \xi_5 (1 + 2\xi_3)], \quad (63)$$

$$Y_4 = 2(1 + 2\xi_3)(1 - \xi_3)(\xi_6 - \xi_5 \xi_6) + 18(1 - \xi_3)\xi_4^2(\xi_6 - \xi_4 \xi_5) - 6(1 + 2\xi_3)(1 - \xi_3)[\xi_5(\xi_6 - \xi_4 \xi_5) + \xi_4(\xi_7 - \xi_5^2)], \quad (64)$$

$$Y_5 = -(1 + 2\xi_3)^2(1 - \xi_3)(\xi_3 \xi_6 - \xi_5 \xi_6) + 3(1 + 2\xi_3)(1 - \xi_3) \times \xi_5(\xi_3 \xi_6 - \xi_4 \xi_5) + 3(1 + 2\xi_3)(1 - \xi_3) \xi_4(\xi_3 \xi_7 - \xi_5^2) - 5(1 + 2\xi_3)(1 - \xi_3)^2(\xi_4 \xi_7 - \xi_5 \xi_6) + 15(1 - \xi_3)^2 \times \xi_4(\xi_4 \xi_6 - \xi_5^2) - 9(1 - \xi_3) \xi_4^2(\xi_3 \xi_6 - \xi_4 \xi_5). \quad (65)$$

One observes that the scattering intensity is fully described by six averages, i.e., ξ_ν , with $\nu = 3-8$. For $K \rightarrow 0$, the previous results [see Eqs. (21) and (22)] are recovered.

V. CONCLUDING REMARKS

The treatment given above is based on the approximate PY integral equation for hard spheres. For one-component HS systems, this approximation gives an exact expression for the term in the density expansion of $g(r)$ and $c(r)$ that is proportional to the particle number density. We expect that this is also true for the multi-component case. This implies, for example, that in the numerator of Eq. (60) terms up to and including the particle density to the power three, and in the denominator to the power two, should be exact. It is generally felt, however, that this approximation is also quite good at higher densities. The remarkable fact is that, for the formulation of the scattering intensity of a mixture of hard spheres with any number of HS components, a closed expression can be obtained containing at the most 18 (averaged) functions of the scattering angle. At any rate, it should give a fair indication of how the scattering of a monodisperse system of spherical particles will change when the system of particles becomes more and more polydisperse. Numerical calculations are planned for a future paper.

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